GT-01

Infinity Learn by Sri Chaitanya Educational Institutions

NEET GRAND TEST - 01 (09-02-2024)

SR ELITE (SET – 1)

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on side-1 and side-2 carefully with blue/black ball point pen only.
2. The test is of 3 hours 20 min duration and Text Booklet contains 200 questions. The test pattern comprises of two sections. Each subject will consist of two sections. Section-A will consist of 35 questions and Section-B will have 15 questions, out of these 15 questions, candidate can chose to attempt any 10 questions.
3. Each question carries 4 marks. For each correct response, the candidate will get 4 marks. For each incorrect response, one mark will be deducted from the total scores. The maximum marks are 720.
4. Use Blue/Black Ball Point Pen only for writing particulars on this page/marking responses.
5. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
6. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
7. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
9. Use of white fluid for correction is NOT permissible on the Answer Sheet.
10. Each candidate must show on demand his/her Admit Card to the Invigilator.
11. No candidate, without special permission of the Superintendent or Invigilator, would leave his/her seat.
12. The candidates should not leave the Examination Hall without handing over there Answer Sheet to the Invigilator on duty and sign the Attendance Sheet twice. Cases where a candidate has not signed the Attendance Sheet second time will be deemed not to have handed over the Answer Sheet and dealt with as an unfair means case.
13. Use of Electronic/Manual Calculator is prohibited.
14. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Hall. All cases of unfair means will be dealt with as per Rules and Regulations of this examination.
15. No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.
16. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.

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Infinity Learn

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THE PERFECT HAT-TRICK
WITH ALL-INDIA RANK 1
IN JEE MAIN 2023
JEE ADVANCED 2023 AND
NEET 2023







BOTANY

SECTION - A

1. Statement I: The breaking of the C – C bonds of complex compounds through oxidation within the cells, leading to release of considerable amount of energy is called respiration.
Statement II: The compounds that are oxidized during this process are known as respiratory substrates.
(1) Both statements are correct
(2) Statement I is correct & II is incorrect
(3) Statement I is incorrect & II is correct
(4) Both statements are incorrect

2. Which of the following is not useful effect of GM plants?
(i) Made crops more tolerant to abiotic stresses likes cold, drought, salt, heat.
(ii) Reduced reliance on chemical pesticides by forming pest-resistant crops.
(iii) Helped to increase post harvest losses.
(iv) Increased mineral uptake by plants so early exhaustion of fertility of soil occurs.
(v) Enhanced nutritional value of food, e.g., Vitamin 'A' enriched rice.
(vi) Create tailor-made plants to supply alternative resources to industries, in the form of starches, fuels and pharmaceuticals.
(1) i, iii and vi (2) iii, iv and v
(3) iii and iv (4) i, ii, iii

3. Match the following columns

Column I	Column II
A. Glyoxysome	i) RuBisCo
B. Chloroplast	ii) Glyoxylate cycle
C. Mitochondria	iii) Photorespiration
D. Peroxisome	iv) Succinate dehydrogenase

(1) A-ii, B-i, C-iv, D-iii
(2) A-ii, B-i, C-iii, D-iv
(3) A-iii, B-i, C-iv, D-ii
(4) A-i, B-ii, C-iv, D- iii

4. Which of the following statements are true for auxin?
(i) They help to prevent fruit and leaf drop at early stages but promote the abscission of older mature leaves and fruits.
(ii) Auxins prevents apical dominance which has its application in tea plantations and hedge-making.
(iii) NAA (naphthalene acetic acid) and 2, 4-D (2, 4-dichlorophenoxyacetic acid) are synthetic auxins.
(iv) Auxins also induce parthenocarpy, e.g., in tomatoes.
(v) 2, 4-D is used to maintain weed-free lawns by gardeners.
(vi) Auxin also controls xylem differentiation and helps in cell division.
(1) i, ii, iii, v and vi
(2) i, ii, iv and vi
(3) i, ii, iii and v
(4) i, iii, iv, v and vi

5. Which one of the following statements about viruses is correct?
(1) Nucleic acid of viruses is known as capsid
(2) Viruses possess their own metabolic system
(3) All viruses contain both RNA and DNA
(4) Viruses are obligate parasites

6. Assertion (A): Mitochondria are the sites of aerobic respiration.
Reason (R): They produce cellular energy in the form of glucose, hence they are called 'power houses' of the cell.
(1) A and R both are true and R is the correct explanation of A
(2) A and R both are true but R is not correct explanation of A
(3) A is true but R is wrong
(4) A and R both are wrong

7. Fixation of one CO_2 molecule through Calvin cycle requires
(1) 1 ATP and 2 NADPH
(2) 2 ATP and 2 NADPH
(3) 3 ATP and 2 NADPH
(4) 2 ATP and 1 NADPH

8. Match the following columns

Column I	Column II
A) Hypodermis in dicot stem	i) Absent
B) Pericycle in dicot stem	ii) Parenchymatous
C) Ground tissue in monocot stem	iii) Collenchymatous
D) Phloem parenchyma in monocot stem	iv) Sclerenchymatous

(1) A - iii, B - iv, C - ii, D – i
(2) A – i, B – ii, C – iv, D – iii
(3) A - iv, B - i, C – iii, D – ii
(4) A – ii, B – iii, C – i, D – iv

9. Match the following list of bioactive substances and their roles.

Bioactive Substance	Role
i) Statin	A. Removal of oil stains
ii) Cyclosporin A	B. Removal of clots from blood vessels
iii) Streptokinase	C. Lowering of blood cholesterol
iv) Lipase	D. Immunosuppressive agent

Choose the correct match.
(1) I – B, II – C, III – A, IV – D
(2) I – D, II – B, III – A, IV – C
(3) I – D, II – A, III – B, IV – C
(4) I – C, II - D, III – B, IV – A

10. Assertion (A): Introns are not found in mature RNA after post transcriptional modifications.
Reason(R): Splicing, occurring after transcription involves removal of introns.
(1) A and R are true and R is the correct explanation of A.
(2) A and R are true and R is not the correct explanation of A.
(3) A is true, R is false.
(4) Both A and R are false.

11. Embryosac is best described as
(1) Progenitor of next generation
(2) Mitotic product from zygote
(3) Female gametophyte
(4) Megasporangium

12. Linkage was discovered by
(1) Mendel
(2) Bateson and Punnett
(3) Sutton (4) Boveri

13. If a cell has 2n number of chromosomes in G_1-phase, what is the number of chromosome in cell after S-phase?
(1) n (2) 4n
(3) 2n (4) 8n

14. In RNAi, the genes are silenced using
(1) ssDNA (2) dsDNA
(3) dsRNA (4) ssRNA

15. Which of the following organisms has chloroplasts with pyrenoids ?
(1) *Chlamydomonas*
(2) *Selaginella*
(3) *Cycas*
(4) *Hibiscus*

16. Assertion (A): Mitosis is equational cell division
Reason(R): During mitosis the daughter cells receive the same number of chromosomes as parent cell.
(1) Both A and R are correct and R is the correct explanation of A
(2) A is correct but R is false
(3) A and R both are false
(4) Both A and R are correct and R is not the correct explanation to A

17. Anaphase of mitosis and anaphase II of meiosis are identical in involving
(i) Chromosomal segregation
(ii) Doubling of chromosome number
(iii) Chromosomal recombinations
(iv) Chromatid segregation
(1) i, ii (2) iii, iv
(3) i, iii (4) ii, iv

18. Non cyclic and cyclic electron transports in chloroplast are similar in involving
(1) PS I
(2) Proton accumulation in lumen
(3) Assimilation of CO_2
(4) 1 and 2

19. Glycolysis does not include
(1) ATP investment
(2) ATP synthesis
(3) Oxidation of triose sugar
(4) CO_2 release

20. How many of the following are natural PGRs occurring in plants ?
IAA, NAA, GA_3, Zeatin, Kinetin, Ethelene, Ethephon, ABA
(1) 3 (2) 7
(3) 5 (4) 6

21. In flowering plants, microsporogenesis and megasporogenesis occur respectively in
(1) Anther and ovule
(2) Pollengrain and megaspore
(3) Ovary and ovule
(4) Anther and pollensac

22. Thalamus develops into false fruit and edible in
(1) Tomato, Banana
(2) Apple, Strawberry
(3) Guava, strawberry, Banana
(4) Mustard, Ground nut, Rice

23. Match the following columns w.r.t probabilities of F_2 progeny of Mendelian dihybrid cross and identify the correct match

Column - I	Column – II
a. Parental phenotypes	i. 1/16
b. Recombinant phenotypes	ii. 10/16
c. Parental dominant phenotype	iii. 9/16
d. Parental recessive phenotype	iv. 6/16

(1) a – ii, b – iv, c – i, d – iii
(2) a – ii, b – iv, c – iii, d – i
(3) a – iv, b –iii, c – ii, d – i
(4) a – iii, b – ii, c – i, d – iv

24. In Avery, Macleod and McCarty experiments transformation of R strain bacteria into S strain was inhibited when ____ was used
(1) RN ase (2) DN ase
(3) Protease (4) 1 and 3

25. Statement–I: Transcriptional unit is DNA flanked on either side by promoter and terminator
Statement–II: Translational unit is mRNA flanked on either side by initiator codon and terminator codon
(1) Both S – I and S – II are correct
(2) Both S – I and S – II are wrong
(3) S – I is correct but S – II is wrong
(4) S – I is wrong but S – II is correct

26. Restriction enzymes are
(1) polymerases (2) Ligases
(3) Lyases (4) Nucleases

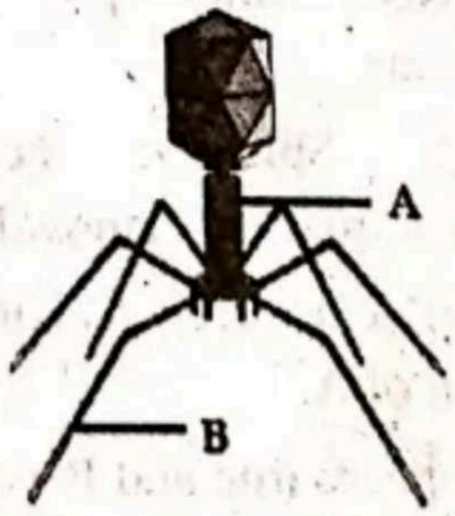
27. Statement–I: The separated bands of DNA are cut out from the agarose gel and extracted from gel piece is called elution.
Statement–II: The normal *E.coli* cells do not carry resistance against kanamycin, tetracycline, ampicillin and chloramphenicol.
(1) Both S-I and S – II are correct
(2) Both S – I and S – II are wrong
(3) S – I is correct but S – II is wrong
(4) S – I is wrong but S – II is correct

28. Match the following and identify the correct match

Column - I	Column – II
a. Golden Rice	i. Gene silencing
b. *Bacillus thuringiensis*	ii. GM rice rich in β - carotene
c. Transposons	iii. Mobile genetic elements
d. RNAi	iv. Biopesticide

(1) a – ii, b – iv, c- iii, d- i
(2) a- i, b – iii, c – iv, d – ii
(3) a – iii, b – ii, c – i, d – iv
(4) a – iv, b – i, c – ii, d – iii

29. Observe the following diagram and identify A and B labelling



(1) Bacteriophage – A –sheath, B – Tail fibrils
(2) Bacteriophage – A – head, B – Tail
(3) Phytophage – A – capsid, B – core
(4) Zoophage – A – collar, B – Tail

30. Which of the following is correct expression ?
(1) Holo enzyme = Apoenzyme + Inhibitor
(2) Holo enzyme = Apoenzyme + cofactor
(3) Coenzyme = Holo enzyme + apoenzyme
(4) Apoenzyme = Holo enzyme + coenzyme

31. In which of the following, F_2 progeny of monohybrid cross show three phenotypes instead of two phenotypes
(A) Dominance
(B) Co dominance
(C) Incomplete dominance
(D) Linkage
(1) A, B (2) C, D
(3) B,C (4) A, D

32. A DNA molecule undergoes replication for 4 generations how many DNA molecules are formed at the end
(1) 12 (2) 8
(3) 32 (4) 16

33. Which of the following is soluble RNA
(1)r-RNA (2) m-RNA
(3) t-RNA (4) SnRNA

34. Which got "transformed" in Griffith experiment?
(1) Mice
(2) R-II bacteria
(3) S-III bacteria
(4) Heat killed S-III bacteria

35. Universal metabolic pathway in all organisms is
(1) Calvin cycle
(2) Glycolysis
(3) Light phase
(4) Fermentation

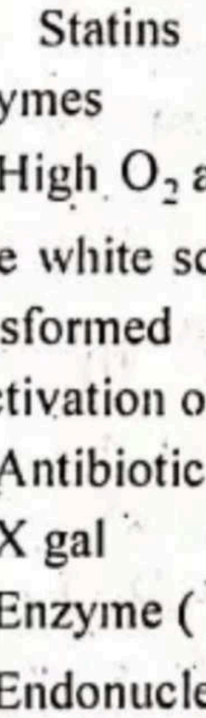
SECTION - B

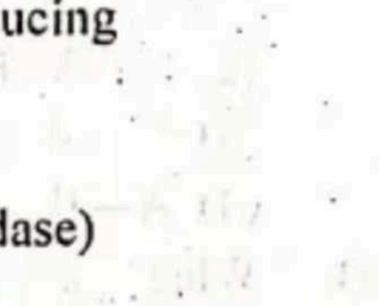
36. Read the following statements
(A) Wall layers in pollen grain
(B) Steps in PCR
(C) Meiospores from a meiocyte
(D) Seeds in mango fruit
Place the above in increasing order with respect to number
(1) A, B, C, D (2)D, A, B, C
(3) D, A, C, B (4) A, D, B, C

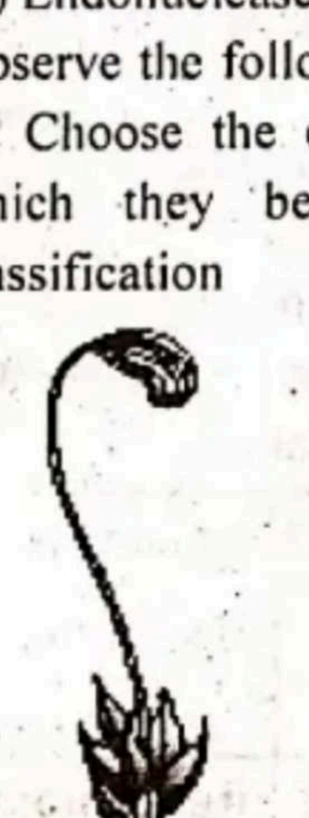
37. Which one among the following is NOT an example for competitive inhibition?
(1) Drugs against bacterial infection
(2) Malate & succinate dehydrogenase
(3) Statins & cholesterol synthesising enzymes
(4) High O_2 and RuBisCo

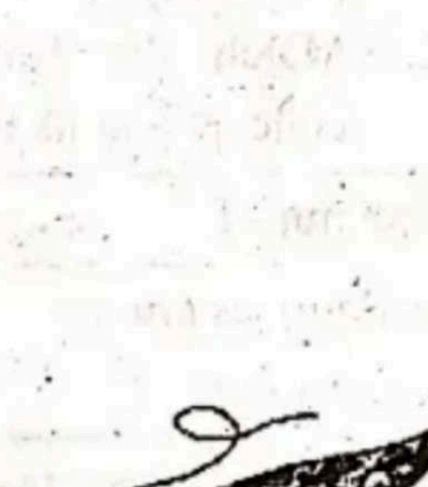
38. Blue white screening method of selection of transformed cells is based on insertional inactivation of a gene producing
(1) Antibiotic
(2) X gal
(3) Enzyme (β - galactosidase)
(4) Endonuclease

39. Observe the following diagrams A, B, C and D. Choose the correct set of kingdoms to which they belong as per Whittaker's classification

A) 

B) 

C) 

D) 

(1) Monera, Protista, Fungi, Plantae
(2) Plantae, Protista, Monera, Fungi
(3) Fungi, Plantae. Monera, Protista
(4) Protista, Fungi, Protista, Monera

40. Match the following

Column - I	Column – II
A)Effect of hard X rays	I) Watson on bacteriophages
B)Embryology in taxonomy	II) Maheswari
C)Origin of species	III) M.S. Swaminathan diversity
D) Production of high yielding wheat varieties	IV) Ernst Mayr

(1) A – I, B – II, C – IV, D - III
(2) A – IV, B – III, C – I, D - II
(3) A – III, B – I, C – II, D - IV
(4) A – II, B – III, C – IV, D – I

41. What is NOT common in between gymnosperms and angiosperms?
(1) Seeds
(2) Dependent gametophytes
(3) Diplontic life cycle
(4) Ploidy of endosperm

42. Match the following in relation to the concept and its explanation

Column - I.	Column – II
a) Gametes are always pure	i) Codominance
b)Heterozygotes resemble both parents	ii)Pleiotropy
c)Physical association of genes on a chromosome	iii) Law of segregation
d) Single gene produces more than one phenotypic effect	iv) Linkage

(1) a – iii, b – i, c – iv, d – ii
(2) a – ii, b – iv, c – iii, d – i
(3) a – iv, b – iii, c - i , d – iii
(4) a – I, b – iv, c – iii, d – ii

43. Read the following statement and choose the correct pair

(A) Bicollateral vascular bundles are observed in anatomy of stem of liliaceae members
(B) Mesophyll exemplify ground tissue in leaves
(C) All three types of simple tissues and two types of complex tissues along with meristem is seen in anatomy of dicot stem
(D) Protoderm represent cork, secondary cortex and cork cambium
(1) B & C (2) B & D
(3) A & D (4) A & C

44. Which are non membrane bound cell organelles observed in an eukaryotic animal cells ?
(A) Ribosomes (B) Inclusion bodies
(C) Centrioles (D) Plastids
(1) A & B (2) B & C
(3) B & D (4) A & C

45. Wind pollination, scutellum, intercalary meristems, bulliform cells, dumbell shaped guard cells, coleoptile and coleorhiza are observed in members of
(1) Lily family (2) Potato family
(3) Bean family (4) Grass family

46.

Microbe utility	Bacteria	Fungi
Household	*Lactobacillus*	A
Bio active molecules	B	*Monascus*
Biopesticide	*Bacillus thuringensis*	C
Bio fertilizer	D	*Glomus*

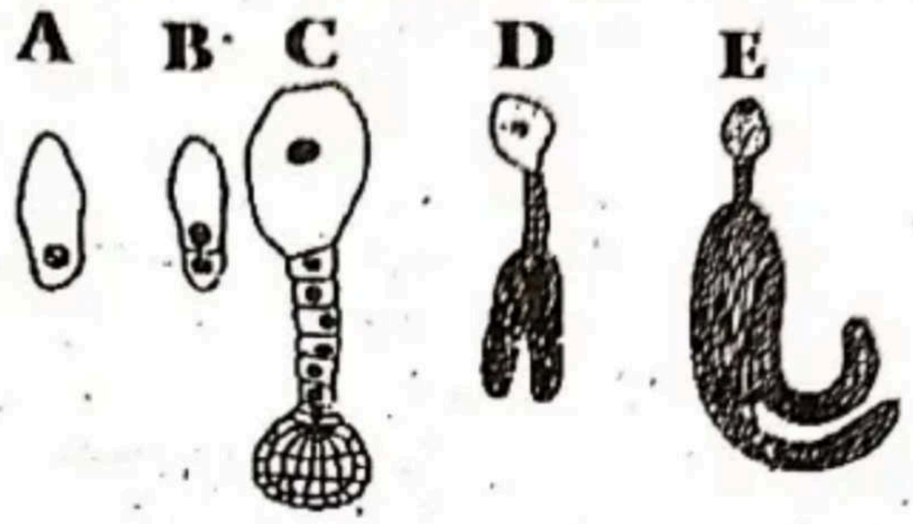
What is A, B, C, D?
(1) A – Yeast, B – *Streptococcus*, C – *Trichoderma*, D – *Rhizobium*
(2) A – *Streptococcus*, B - Yeast C - *Rhizobium*, D – *Trichoderma*
(3) A – *Trichoderma*, B – *Rhizobium*, C – *Streptococcus*, D - Yeast
(4) A – *Rhizobium*, B - *Trichoderma*, C – Yeast, D - *Streptococcus*

47. A unit composed of A_ and _B_ linked by _C_ is a nucleoside. What is A, B, & C ?
(1) A – Sugar, B - Phosphoric acid
C - Phospho ester bond
(2) A – Sugar, B - Nitrogen base
C - Glycosidic bond
(3) A – Nitrogen base, B - Phosphoric acid , C - Phosphoric acid
(4) A – Phosphoric, B - Nitrogen base, C - Hydrogen bond acid

48. Which of the following is incorrect way to introduce alien DNA into host cells ?
(i)Bacteria is made 'competent' to take up DNA by treating them with a specific concentration of a divalent cation, such as calcium.
(ii) Recombinant DNA is transfered into competent bacteria by incubating the cells with recombinant DNA on ice, heat shock, and then putting them back on ice.
(iii) Recombinant DNA is directly injected into the nucleus of plant cell by micro-injection.
(iv) Recombinant DNA is directly injected into the nucleus of an animal cell by biolistics or gene gun.
(v) 'Disarmed Pathogen' vectors, when allowed to infect the cell, transfer the recombinant DNA into the host.
(1) i and ii (2) ii and iii
(3) iii and iv. (4) iv and v

49. Two carbon compound formed in chloroplast during photorespiration is
(1) Glycine
(2) 2-phosphoglycolate
(3) 3PGA
(4) PEP

50. The figure below shows five steps (A, B, C, D, E) of embryo development.
Select the option giving correct identification together with what is represents ?



Options
(1) C-Represents mature stage of embryo
(2) D-Formed by meiosis in zygote
(3) C-Represents globular embryo
(4) A- Zygote formed by triple fusion

ZOOLOGY

SECTION – A

51. Notochord is persistent throughout life in
(1) *Hippocampus* (2) *Scoliodon*
(3) *Salamandra* (4) *Ascidia*

52. LH surge induces
(1) menstruation (2) parturition
(3) fertilization (4) ovulation

53. Statement – I : Adhering junctions facilitate the cells to communicate with each other by connecting cytoplasm of adjoining cells, for rapid transfer of ions.
Statement – II : Gap junctions help to stop substances from leaking across a tissue.
(1) Both statement I and statement II are true
(2) Both statements I and statement II are false
(3) Statement I is true but statement II is false
(4) Statement I is false but statement II is true

54. Which of the following is not the feature of simple columnar epithelium ?
(1) Flattened cells with irregular boundaries
(2) Nuclei are located at the base of cells
(3) Free surface may have microvilli
(4) Found in the lining of stomach

55. Which of the following occurs during expiration in humans ?
(1) Contraction of diaphragm
(2) Increase in the volume of thoracic chamber
(3) Relaxation of external inter – coastal muscles
(4) Decréase in the intra – pulmonary pressure

56. Which of the following contraceptives are very similar to each other in their mechanism of action ?
(1) Saheli and cervical cap
(2) LNG -20 and progestasert
(3) Nirodh and Cu-7
(4) Vault and Vasectomy

57. Assertion – (A) : The number of action potentials that could be generated in a minute vary at different parts of the nodal system
Reason (R) : The nodal musculature has the ability to generate action potentials without any external stimuli
(1) Both 'A' and 'R' are true and 'R' is not the correct explanation of 'A'.
(2) 'A' is false, but 'R' is true
(3) 'A' is true and 'R' is false
(4) Both 'A' and 'R' are true and 'R' is the correct explanation of 'A'.

58. Sickle cell anaemia is controlled by a single pair of allele
(1) Hb^A and Hb^B
(2) Hb^{A1} and Hb^{A2}
(3) Hb^A and Hb^S
(4) Hb^B and Hb^{A1}

59. The cells of Bowman's capsule that wrap around the glomerular capillaries are
(1) Podocytes (2) Fibrocytes
(3) Adipocytes (4) Chondrocytes

60. Which of the following drug interferes with the transport of dopamine and has a potent stimulating action on CNS ?
(1) Heroin (2) Barbiturates
(3) Morphine (4) Cocaine

61. Which of the following is the bone of pectoral girdle ?
(1) Clavicle (2) Occipital
(3) Carpal (4) Zygomatic

62. Choose the incorrect statement about predators
(1) Acts as conduits for energy transfer across trophic levels.
(2) Phytophagous insects are considered as herbivores but not as predators
(3) Help in maintaining species diversity in a community.
(4) Biologoical agricultural pest control is based on the ability of the predator to regulate prey population

63. Divergent evolution explains about
(1) different structures evolving for the same functions
(2) similar structures evolving for the different functions
(3) different structures evolving for the different functions
(4) similar structures evolving for the same functions

64. Which of the following ecological pyramid is generally spindle shaped ?
(1) Pyramid of biomass in a sea
(2) Pyramid of number in a tree ecosystem
(3) Pyramid of energy in a grassland
(4) Pyramid biomass in a forest

65. An adrenal hormone responsible for the regulation of blood glucose level is
(1) Insulin (2) Thyroxine
(3) Glucagon (4) Cortisol

66. Statement – I : A stable community must be either resistant or resilient to occasional disturbances.
Statement – II : Tilman found that plots with more species showed too much year to year variation in total biomass.
(1) Both statement I and statement II are true
(2) Both statements I and statement II are false
(3) Statement I is true but statement II is false
(4) Statement I is false but statement II is true

67. The mantle cavity possess feather like gills in
(1) arthropods (2) annelids
(3) molluscs (4) echinoderms

68. Which of the following is incorrect ?
(1) Protooncogenes are present in normal cells
(2) Vaccines generate memory – B and memory - T cells
(3) Anti-histamine, adrenaline and steroids quickly reduce the symptoms immunodeficiency
(4) Rheumatoid arthritis is an autoimmune disease

69. Match the list I with List II

	List I		List II
A	Ossicles	i	*Aplysia*
B	Radula	ii	*Nereis*
C	Osculum	iii	*Asterias*
D	Segments	iv	*Euspongia*

	A	B	C	D
(1)	iii	i	ii	iv
(2)	iv	i	iii	ii
(3)	iii	i	iv	ii
(4)	iv	i	ii	iii

70. Eli Lilly an American company prepared two DNA sequences corresponding to A and B chains of human insulin and introduced them in
(1) cosmid of *Helicobactor*
(2) Plasmid of *E.coli*
(3) Cosmid of *Yeast*
(4) Plasmid of *Bacteriophage*

71. In frog, all of the following are cellular aggregations around nerve endings, except
(1) Sensory papillae
(2) Internal ear
(3) Taste buds
(4) Nasal epithelium

72. Which of the following represents the correct equation for logistic growth model ?
(1) $\frac{dN}{dt} = rN(1-\frac{N}{K})$
(2) $N_t = N_0\, e^{rt}$
(3) $\frac{dN}{dt} = rN$
(4) $\frac{dN}{dt} = rN\left(\frac{N-K}{N}\right)$

73. Which of the following is an enzyme complex required for the conversion of prothrombin into thrombin ?
(1) Streptokinase
(2) Thrombokinase
(3) Carbonic anhydrase
(4) Carboxypeptidose

74. Assertion (A) : Placenta is a structural and functional unit between developing embryo (foetus) and maternal body.
Reason (R) : It facilitate the supply of oxygen and nutrients to the embryo and also removal of carbon dioxide and excretory materials produced by the embryo.
(1) Both 'A' and 'R' are true and 'R' is the correct explanation of 'A'.
(2) 'A' is false, but 'R' is true
(3) 'A' is true and 'R' is false
(4) Both 'A' and 'R' are true and 'R' is not the correct explanation of 'A'.

75. Select the incorrect statement from the following
(1) A fall in GFR activates JG cells to release angiotensinogen
(2) Angiotensin II is a vasoconstrictor and increases blood pressure
(3) Atrial natriuretic factor is a vasodilator
(4) ADH facilitates water reabsorption from the distal tubules of nephron

76. Assertion (A) : Hepatitis B and HIV infections are not completely curable.
Reason (R) : HIV and HBV can be transmitted by sharing of injection needles and surgical instruments with infected persons.
(1) Both 'A' and 'R' are true and 'R' is the correct explanation of 'A'.
(2) 'A' is false, but 'R' is true
(3) 'A' is true and 'R' is false
(4) Both 'A' and 'R' are true and 'R' is not the correct explanation of 'A'.

77. Each muscle fibre of a skeletal muscle is lined by the plasma membrane called
(1) Fascia (2) Myelin sheath
(3) Perimycium (4) Sacrolemma

78. Which of the following part of nephron is impermeable to water ?
(1) Ascending limb
(2) Collecting duct
(3) Proximal convoluted tubule
(4) Distal convoluted tubule

79. Which of the following represents the action of glucagon ?
(1) It increases blood glucose level by stimulating glycogenesis.
(2) It decreases blood glucose level by stimulating glycogenolysis.
(3) It increases blood glucose level by stimulating gluconeogenesis.
(4) It decreases blood glucose level by stimulating proteolysis.

80. Match the list I list II

	List I		List II
A	IC	i	VC + RV
B	VC	ii	TV + IRV
C	FRC	iii	TV + IRV+ERV
D	TLC	iv	RV + ERV

	A	B	C	D
(1)	ii	iii	i	iv
(2)	iv	iii	i	ii
(3)	iv	iii	ii	i
(4)	ii	iii	iv	i

81. The blastomeres in the blastocyst are arranged into an outer layer called
(1) trophoblast (2) inner cell mass
(3) granulosa (4) fibroblast

82. Match the list I with List II regarding reproductive system of *Periplaneta americana.*

	List I		List II
A	Mushroom shaped gland	i	$6^{th} - 7^{th}$
B	Spermathecae	ii	6th
C	Testes	iii	$4^{th} - 6^{th}$
D	Ovaries	iv	$2^{nd} - 6^{th}$

	A	B	C	D
(1)	i	ii	iii	iv
(2)	i	ii	iv	iii
(3)	ii	i	iii	iv
(4)	ii	i	iv	iii

83. Which of the following ARTs involves the transfer of ovum into the fallopian tube ?
(1) ICSI (2) IVF
(3) ZIFT (4) GIFT

84. If sodium – potassium pump stops working, it results in
(1) increase in the Na^+ inside the axon
(2) decrease in the K^+ outside the axon
(3) increase in the Na^+ outside the axon
(4) increase in the K^+ inside the axon

85. Which of the following is the correct match ?
(1) Males are heterogametic – Honeybee
(2) Females are homogametic – Butterfly
(3) Males are homogametic – Fruitfly
(4) Females are heterogametic – Pigeon

SECTION – B

86. Which of the following members are all marine ?
(1) Echinoderms, Cnidarians and protochordates
(2) Protochordates, Chondrichthyes and Annelids
(3) Hemichordates, Echinoderms and Arthropods
(4) Ctenophores, Echinoderms and Hemichordates

87. Which of the following is the correct sequence of parts in the foregut of cockroach ?
(1) Pharynx – Oesophagus – Gizzard – Crop
(2) Pharynx – Oesophagus – Crop – Gizzard
(3) Ileum – Colon – Rectum – Mesenteron
(4) Mesenteron – Ileum – Colon – Rectum

88. In frog, formation of chyme takes place in the
(1) Stomach
(2) Small intestine
(3) Rectum
(4) Oesophagus

89. Match the Column I with Column II

	List I		List II
A	*Homo sapiens*	i	Hunted with stone weapons
B	*Homo erectus*	ii	Buried their dead
C	*Australopithecines*	iii	Probably ate meat
D	Neanderthal man	iv	Pre-historic cave art

	A	B	C	D
(1)	iv	iii	i	ii
(2)	iii	iv	ii	i
(3)	ii	i	iii	iv
(4)	i	ii	iv	iii

90. Partial pressure of oxygen in the systemic arteries is
(1) less than that of alveolus
(2) greater than that of pulmonary vein
(3) less than that of systemic vein
(4) equal to that of tissues

91. Which of the following hormones are produced by pars distalis ?
a) Prolactin
b) Follicle Stimulating Hormone
c) Somatostatin
d) Gonadotropin Releasing Hormone
e) Melanocyte Stimulating Hormone
(1) a, b and e only (2) b, d and e only
(3) b and d only (4) a and b only

92. The state of heart when it is not pumping blood effectively enough to meet the needs of the body is called
(1) Cardiac arrest (2) Heart attack
(3) Atherosclerosis (4) Heart failure

93. Statement – I : Midbrain is located between the thalamus/ hypothalamus of hindbrain and pons of the forebrain.
Statement – II : Three major regions make up the brain stem are midbrain, pons and cerebellum.
(1) Both statement I and statement II are true
(2) Both statements I and statement II are false
(3) Statement I is true but statement II is false
(4) Statement I is false but statement II is true

94. Assertion (A) : Pre-synaptic neuron transmits an impulse across the synaptic cleft to the post synaptic neuron in a chemical synapse.
Reason (R) : In a chemical synapse, the membranes of pre and post synaptic neurons are in very close proximity.
(1) Both 'A' and 'R' are true and 'R' is the correct explanation of 'A'.
(2) 'A' is false, but 'R' is true
(3) 'A' is true and 'R' is false
(4) Both 'A' and 'R' are true and 'R' is not the correct explanation of 'A'.

95. According to Oparin and Haldane, the first form of life could have come from
(1) decaying and rotting matter like straw
(2) pre-existing living organisms
(3) resistant spores of other planets
(4) pre – existing organic molecules

96. The technique of DNA fingerprinting was initially developed by
(1) Frederick Sanger
(2) Mathew Messelson
(3) Alec Jefferys
(4) Francis Crick

97. Match the following List I with List II

	List I		List II
A	*Entamoeba histolytica*	i	Large intestine
B	*Wuchereria malayi*	ii	Lymphatic vessels
C	*Haemophilus influenzae*	iii	Alveoli of lungs
D	*Plasmodium vivax*	iv	Red blood cells

	A	B	C	D
(1)	i	ii	iv	iii
(2)	i	ii	iii	iv
(3)	ii	i	iv	iii
(4)	ii	i	iii	iv

98. Between which among the following, the relationship is not an example of commensalism ?
(1) Orchid and the tree on which it grows
(2) Cattle egret and grazing cattle
(3) Sea anemone and clown fish
(4) Female wasp and fig species

99. Which of the following is the correct set of conservation approach and an example ?
(1) In situ – Wildlife safari park
(2) Ex situ – National park
(3) In situ – Wild life sanctuary
(4) Ex situ – Sacred grove

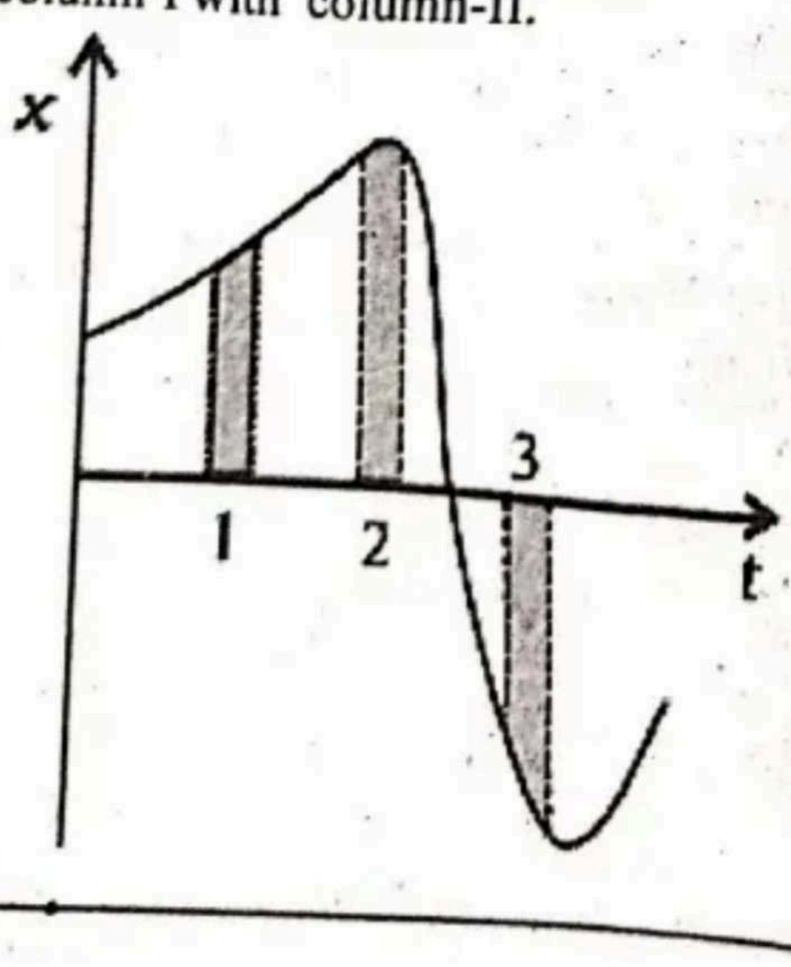
100. The process of breakdown of detritus into inorganic nutrients by the action of bacterial and fungal enzymes is
(1) Fragmentation
(2) Leaching
(3) Mineralisation
(4) Catabolism

PHYSICS

SECTION – A

101. A force F is given by $F = at + bt^2$, where t is time. What are the dimensions of a and b
(1) MLT^{-3} and ML^2T^{-4}
(2) MLT^{-3} and MLT^{-4}
(3) MLT^{-1} and MLT^{0}
(4) MLT^{-4} and MLT^{1}

102. Figure gives the x-t plot of a particle in one-dimensional motion. Three different equal intervals of time are shown. From the graph, match from column-I with column-II.



column-I
(A) In which interval is the average speed greatest ?
(B) In which interval is the average speed least?
(C) the sign of average velocity in interval (1)
(D) the sign of average velocity in interval (3)

column-II.
(P) interval (3)
(Q) interval (2)
(R) positive
(S) negative

(1) A-P ; B-Q ; C-S ; D-R
(2) A-S ; B-R ; C-Q ; D-P
(3) A-P ; B-Q ; C-R ; D -S
(4) A-Q ; B-P ; C-R ; D-S

103. The speed of a swimmer in still water is 20 m/s. The speed of river water is 10 m/s and is flowing due east. If he is standing on the south bank and wishes to cross the river along the shortest path, the angle at which he should make his strokes w.r.t. north is given by
(1) 30^0 west (2) 0^0
(3) 60^0 west (4) 45^0 west

104. A brick of mass 2 kg is sliding down with constant velocity on a plane inclined at an angle of 45^0 with the horizontal. The force of friction will be
(1) $19.6 \sin 45^0$ (2) $19.6 \tan 45^0$
(3) $9.8 \sin 45^0$ (4) $9.8 \cos 45^0$

105. A particle moves in the xy-plane under the action of a force F such that the components of its linear momentum p at any time t are $p_x = 2\cos t$, $p_y = 2\sin t$. The angle between Force F and momentum p at time t is
(1) 90^0 (2) 0^0
(3) 180^0 (4) 30^0

106. In an elastic collision of two particles, the following is conserved
(1) Momentum of each particle
(2) Speed of each particle
(3) Kinetic energy of each particle
(4) Total kinetic energy of both the particles

107. An inclined track ends in a circular loop of diameter D. From what height on the track, a particle should be released so that it completes that loop in the vertical plane ? (All are smooth surfaces)

(1) $\frac{5D}{2}$ (2) $\frac{5D}{4}$

(3) $\frac{2D}{5}$ (4) $\frac{4D}{5}$

108. Moment of inertia of a uniform circular disc about a diameter is I. Its moment of inertia about an axis perpendicular to its plane and passing through a point on its rim will be

(1) 5I (2) 6I

(3) 3I (4) 4I

109. The position of a particle is given by : $\vec{r} = (\hat{i} + 2\hat{j} - \hat{k})$ and momentum $\vec{P} = (3\hat{i} + 4\hat{j} - 2\hat{k})$. The angular momentum is perpendicular to

(1) X-axis

(2) Y-axis

(3) Z-axis

(4) Line at equal angles to all the three axes

110. Statement(A) : For Geostationary satellites, height above the earth is about 35,800 km.

Statement(B):For Polar satellites, height above the earth is about 500 to 800 km.

Statement(C): For Geostationary satellites, T = 24 hours.

Statement(D):For Polar satellites, time period is around 100 minutes.

(1) A and B are true

(2) A ,B and C are true

(3) A, C and D are true

(4) all are true

111. The maximum velocity of an electron emitted by light of wavelength λ incident on the surface of a metal of work function ϕ, is

(1) $\left[\frac{2(hc+\lambda\phi)}{m\lambda}\right]^{1/2}$ (2) $\frac{2(hc-\lambda\phi)}{m}$

(3) $\left[\frac{2(hc-\lambda\phi)}{m\lambda}\right]^{1/2}$ (4) $\left[\frac{2(h\lambda-\phi)}{m}\right]^{1/2}$

112. If the wavelength of the first line of the Balmer series of hydrogen is 6561 $\AA$, the wavelength of the second line of the series should be

(1) 13122 $\AA$ (2) 3280 $\AA$

(3) 4860 $\AA$ (4) 2187 $\AA$

113. One requires energy E_n to remove a nucleon from a nucleus and an energy E_e to remove an electron from the orbit of an atom. Then

(1) $E_n = E_e$ (2) $E_n < E_e$

(3) $E_n > E_e$ (4) $E_n \geq E_e$

114. The width of forbidden gap in silicon crystal is 1.1 eV. When the crystal is converted in to a N-type semiconductor the distance of Fermi level from conduction band is

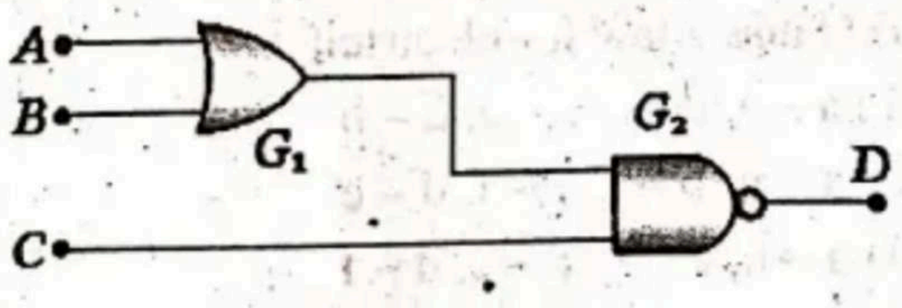
(1) Greater than 0.55 eV

(2) Equal to 0.55 eV

(3) Lesser than 0.55 eV

(4) Equal to 1.1 eV

115. For the given combination of gates, if the logic states of inputs A, B, C are as follows A = B = C = 0 and A = B = 1, C = 0 then the logic states of output D are



(1) 0, 0 (2) 0, 1

(3) 1, 0 (4) 1, 1

116. Statement(A) : The Sun is visible a little before the actual sunrise and until a little after the actual sunset due to refraction of light through the atmosphere

Statement(B) : Mass density of 'Turpentine' is less than that of water but its optical density is higher

(1) only A is true

(2) only B is true

(3) both A and B are true

(4) both A and B are false

117. When light rays are incident on a prism at an angle of 45^0, the minimum deviation is obtained. If refractive index of the material of prism is $\sqrt{2}$, then the angle of prism will be

(1) 30^0 (2) 40^0

(3) 50^0 (4) 60^0

118. If two light waves having same frequency have intensity ratio 4 : 1 and they interfere, the ratio of maximum to minimum intensity in the pattern will be

(1) 9 : 1 (2) 3 : 1

(3) 25 : 9 (4) 16 : 25

119. Match list-I with list-II.

List-I

a) $\oint \vec{E}.\overrightarrow{dA} = \frac{Q_{in}}{\epsilon_0}$

b) $\oint \vec{B}.\overrightarrow{dA} = 0$

c) $\oint \vec{E}.\overrightarrow{d\ell} = \frac{-d\phi_B}{dt}$

d) $\oint \vec{B}.\overrightarrow{d\ell} = \mu_0 i_c + \mu_0 \epsilon_0 \frac{d\phi_E}{dt}$

List-II

e) Ampere-maxwell law

f) Faraday's law

g) Gauss's law of magnetism

h) Gauss's law for electricity

(1) a – e, b – f, c – g, d – h

(2) a – h, b – g, c – f, d – e

(3) a – h, b – g, c – e, d – f

(4) a – g, b – h, c – f, d – e

120. A boat is moving due east in a region where the earth's magnetic field is $5.0\times10^{-5} NA^{-1}m^{-1}$ due north and horizontal. The boat carried a vertical aerial 2m long. If the speed of the boat is $1.50ms^{-1}$, the magnitude of the induced emf in the wire of aerial is:

(1) 0.75 mV (2) 0.50 mV

(3) 0.15 mV (4) 1 mV

121. When you walk through a metal detector carrying a metal object in your pocket, it raises an alarm. This phenomenon works on

(1) Electromagnetic induction.

(2) Resonance in ac circuits.

(3) Mutual induction in ac circuits.

(4) Interference of electromagnetic waves.

122. Assertion(A): In an uniform magnetic field, speed and kinetic energy remains the same for a moving charged particle.

Reason (R): In the case of motion of a charge in a magnetic field, the magnetic force is perpendicular to the velocity of the particle. So, no work is done and no change in the magnitude of the velocity (speed) and kinetic energy is produced.

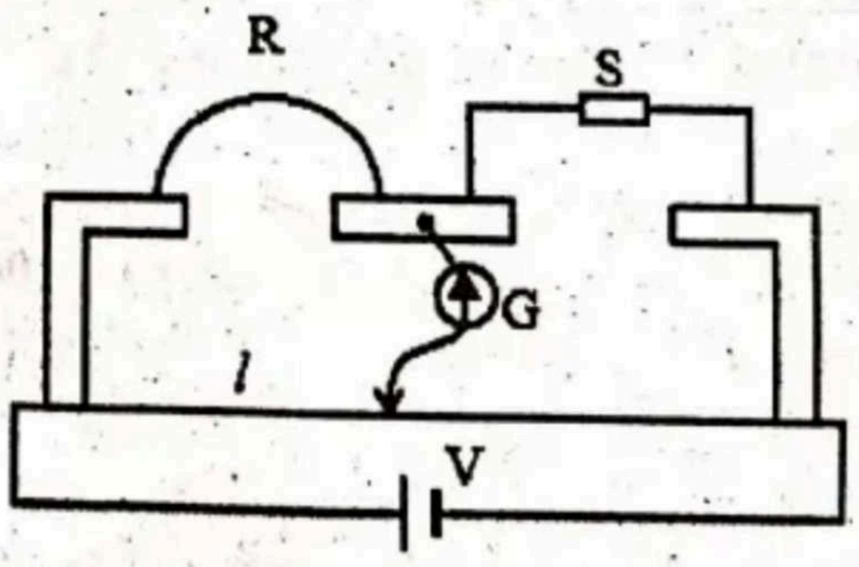
(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true and R is not the correct explanation of A

(3) A is true but R is false

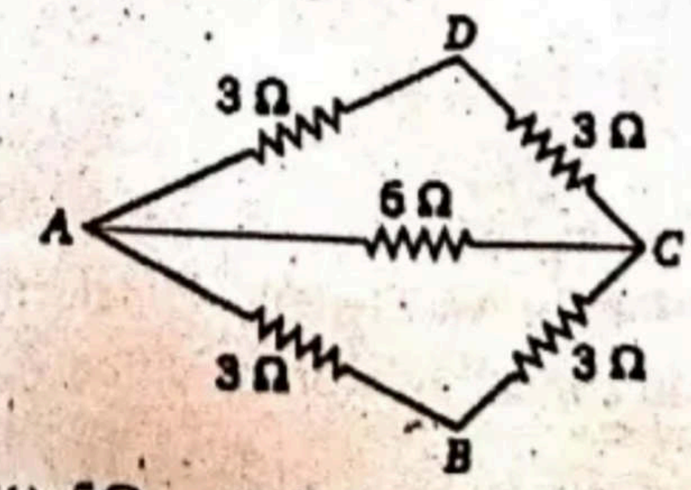
(4) Both A and R are false

123. In a meter bridge experiment S is a standard resistance. R is a resistance wire. It is found that balancing length is l= 25 cm. If R is replaced by a wire of half length and half diameter that of R of same material, then the balancing distance 'l' (in cm) will now be



(1) 10 (2) 20 (3) 30 (4) 40

124. The effective resistance between the points A and B in the figure is



(1) 5Ω (2) 2Ω (3) 3 Ω (4) 4 Ω

125. If the potential function is given by $V = 4x + 3y$, then the magnitude of electric field intensity at the point (2, 1) will be.

(1) 11 (2) 5 (3) 7 (4) 1

126. A parallel plate air capacitor is charged to a potential difference of V volts. After disconnecting the charging battery, the distance between the plates of the capacitor is increased using an insulating handle. As a result, the potential difference between the plates

(1) does not change (2) becomes zero

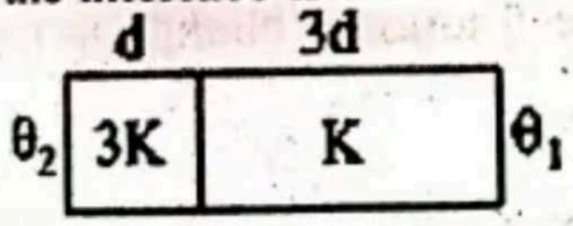
(3) increases (4) decreases

127. A charge 'q' is placed at the centre of the line joining two equal charges 'Q'. The system of the three charges will be in equilibrium if 'q' is equal to

(1) Q/2 (2) – Q/4

(3) Q/4 (4) – Q/2

128. Two materials having coefficients of thermal conductivity '3K' and 'K' and thickness 'd' and '3d' respectively, are joined to form a slab as shown in the figure. The temperatures of the outer surfaces are $'\theta_2'$ and $'\theta_1'$ respectively, $(\theta_2 > \theta_1)$. The temperature at the interface is :



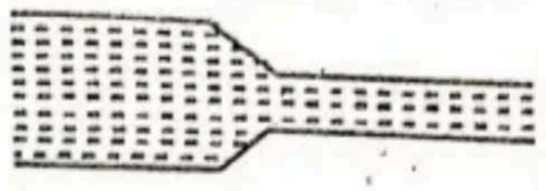
(1) $\frac{\theta_1}{10} + \frac{9\theta_2}{10}$ (2) $\frac{\theta_2 + \theta_1}{2}$

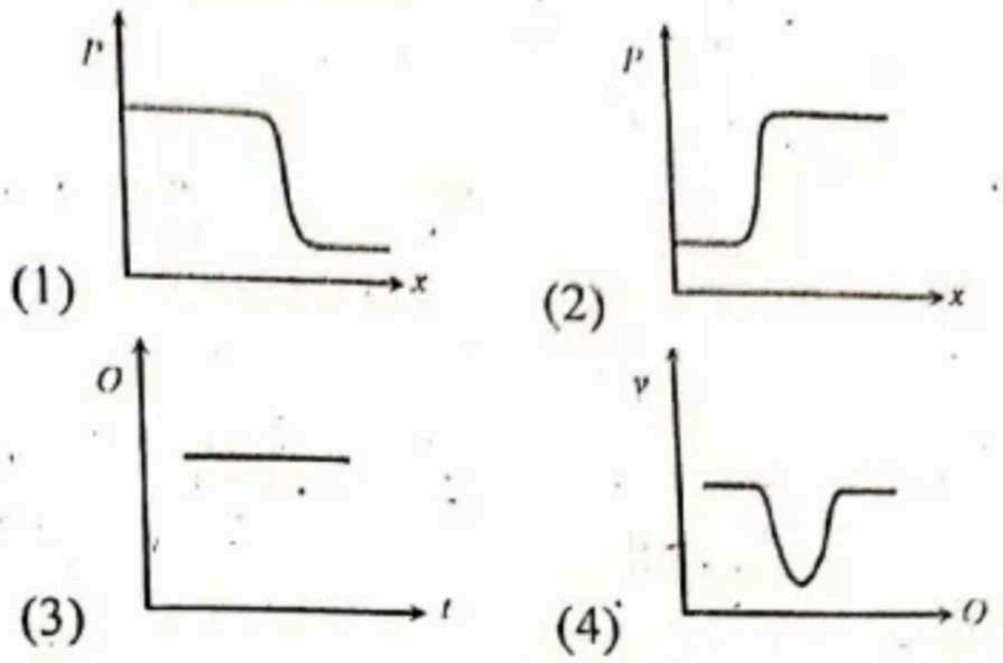
(3) $\frac{\theta_1}{6} + \frac{5\theta_1}{6}$ (4) $\frac{\theta_1}{3} + \frac{2\theta_2}{3}$

129. If two rods of length L and 2L having coefficients of linear expansions α and 2α respectively are connected so that total length becomes 3L, the average coefficient of linear expansion of the composite rod equals to

(1) $\frac{3}{2}\alpha$ (2) $\frac{5}{2}\alpha$ (3) $\frac{5}{3}\alpha$ (4) α

130. Water flows through a frictionless duct with a cross-section varying as shown in fig. Pressure p at points along the axis is represented by





131. Maximum velocity in SHM is v_m. The average velocity during motion from one extreme point to the other extreme point will be

(1) $\frac{\pi}{2}v_m$ (2) $\frac{2}{\pi}v_m$

(3) $\frac{4}{\pi}v_m$ (4) $\frac{\pi}{4}v_m$

132. The maximum load a wire can withstand without breaking, when its length is reduced to half of its original length, will

(1) Be double. (2) Be half.

(3) Be four times. (4) Remain same.

133. The value of $\gamma\left(=\frac{C_p}{C_v}\right)$, for hydrogen helium and another ideal diatomic gas X (Whose molecules are not rigid but have an additional vibrational mode), are respectively equal to

(1) $\frac{7}{5}, \frac{5}{3}, \frac{9}{7}$ (2) $\frac{5}{3}, \frac{7}{5}, \frac{9}{7}$

(3) $\frac{5}{3}, \frac{7}{5}, \frac{7}{5}$ (4) $\frac{7}{5}, \frac{5}{3}, \frac{7}{5}$

134. At pressure P and absolute temperature T a mass m of an ideal gas fills a closed container of volume V. An additional mass 2m of the same gas is added into the container and the volume is then reduced to $\frac{V}{3}$ and the temperature to $\frac{T}{3}$. The pressure of the gas will now be:

(1) $\frac{P}{3}$ (2) P (3) 3P (4) 9P

135. STATEMENT(A): A quasi-static process is an infinitely slow process such that the system remains in thermal and mechanical equilibrium with the surroundings throughout.
STATEMENT(B): Spontaneous processes of nature are irreversible.

(1) A is true and B is false

(2) *A is false and B is true*

(3) *Both A and B are true*

(4) *Both A and B are false*

SECTION - B

136. A car starts from rest and accelerates at 5 m/s^2. At t = 4 s, a ball is dropped out of a window by a person sitting in the car. What is the velocity and acceleration of the ball in air at t = 6s? (Neglect air resistance and Take $g = 10\ m/s^2$)

(1) 20 m/s, 10 m/s^2 (2) $20\sqrt{2}$ m/s, 0

(3) $20\sqrt{2}$ m/s, 10 m/s^2 (4) 20 m/s, 5 m/s^2

137. Assertion(A) : A bullet is fired from a rifle. If the rifle recoils freely, the kinetic energy of rifle is less than that of the bullet
Reason(R) : In the case of rifle-bullet system, the law of conservation of Energy is not valid.

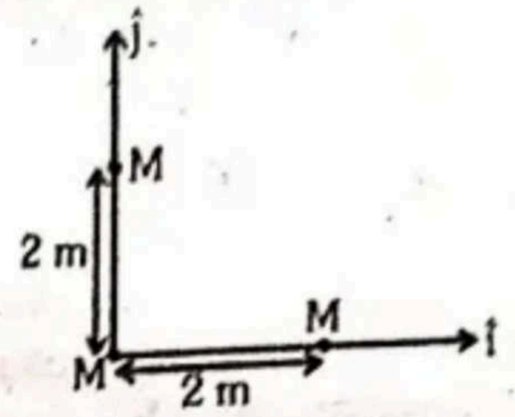
(1) A and R are true and R is the correct explanation of A

(2) A and R are true and R is not the correct explanation of A

(3) A is true and R is false

(4) A is false and R is false

138. Three identical spheres, each of mass M, are placed at the corners of a right angle triangle with mutually perpendicular sides equal to 2 m (see figure). Taking the point of intersection of the two mutually perpendicular sides as the origin, find the position vector of centre of mass



(1) $2(\hat{i}+\hat{j})$ (2) $(\hat{i}+\hat{j})$

(3) $\frac{2}{3}(\hat{i}+\hat{j})$ (4) $\frac{4}{3}(\hat{i}+\hat{j})$

139. A photon and an electron have equal energy E. Then $(\lambda_{photon}/\lambda_{electron})$ is proportional to

(1) $\sqrt{E}$

(2) $1/\sqrt{E}$

(3) $1/E$

(4) Does not depend upon E

140. Assertion(A) : Photodiodes are preferably used in the reverse bias condition for measuring light intensity.
Reason(R) : The fractional change due to the photo-effects on the minority carrier dominated reverse bias current is more easily measurable than the fractional change in the forward bias current.

(1) A and R are true and R is the correct explanation of A

(2) A and R are true and R is not the correct explanation of A

(3) A is true and R is false

(4) A is false and R is true

141. Four convergent lenses have focal lengths 100 cm, 10 cm, 4 cm and 0.3 cm. For a telescope with maximum possible magnification, we choose the lenses of focal length

(1) 100 cm, 0.3 cm

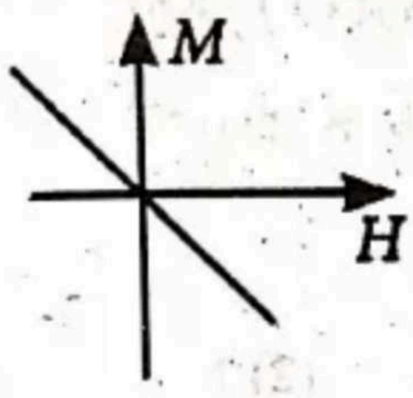
(2) 10 cm; 0.3 cm

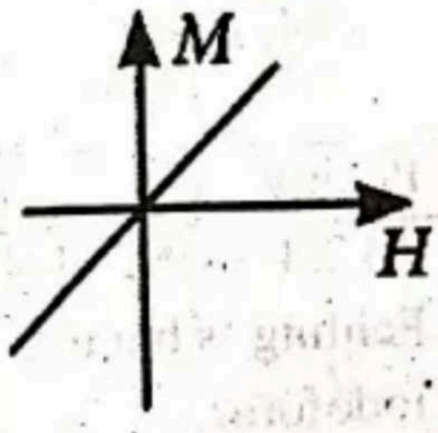
(3) 10 cm, 4 cm

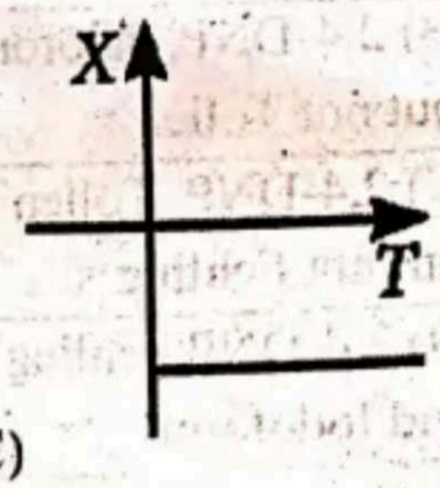
(4) 100 cm, 4 cm

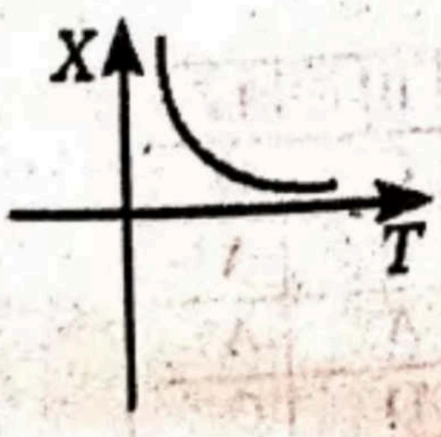
142. The wavelength of light observed on the earth, from a moving star is found to decrease by 0.05%. Relative to the earth the star is
(1) Moving away with a velocity of $1.5\times10^5\,m/s$
(2) Coming closer with a velocity of $1.5\times10^5\,m/s$
(3) Moving away with a velocity of $1.5\times10^4\,m/s$
(4) Coming closer with a velocity of $1.5\times10^4\,m/s$

143. Following plots show magnetization (M) vs magnetizing field (H) and magnetic susceptibility (χ) vs temperature (T) graph

A)


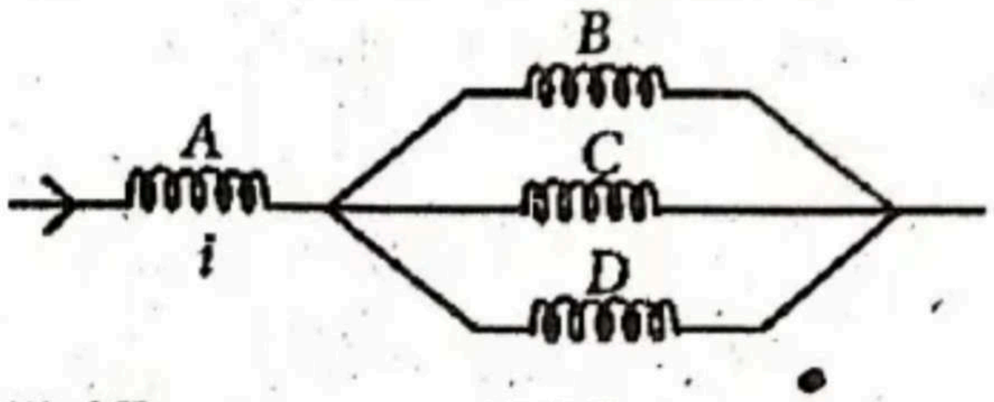
B)


C)


D)


Which of the following combination will be represented by a diamagnetic material?
(1) (B), (C) (2) (B), (D)
(3) (A), (D) (4) (A), (C)

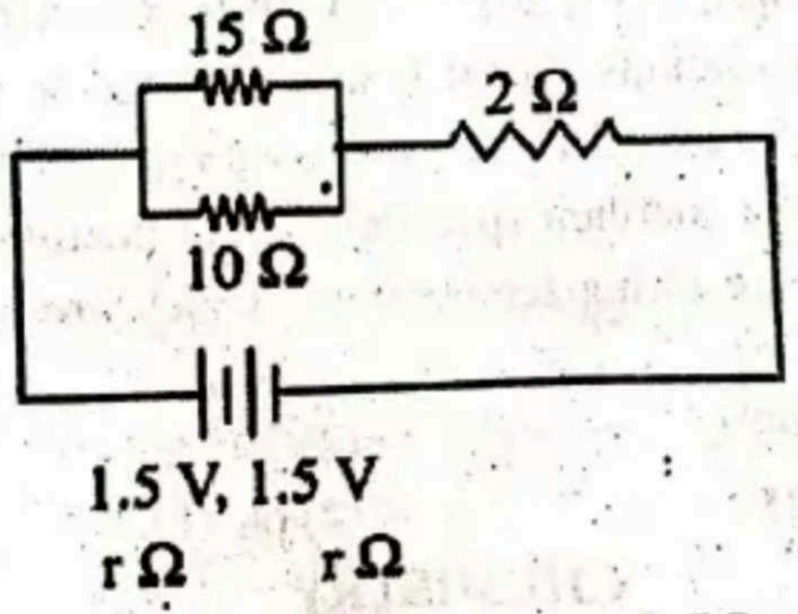
144. Four identical long solenoids A, B, C and D are connected to each other as shown in the figure. If the magnetic field at the center of A is 3 T, the field at the center of C would be (Assume that the magnetic field is confined with in the volume of respective solenoid)



(1) 6 T (2) 9 T
(3) 1 T (4) 12 T

145. A tuning fork of frequency 480 Hz is used in an experiment for measuring speed of sound (v) in air by resonance tube method. Resonance is observed to occur at two successive lengths of the air column, l_1=30 cm and l_2= 70cm. Then, v is equal to :
(1) 332 ms^{-1} (2) 384 ms^{-1}
(3) 338 ms^{-1} (4) 379 ms^{-1}

146. In the given circuit, an ideal voltmeter connected across the 10Ω resistance reads 2V. The internal resistance r, of each cell is :



(1) 1Ω (2) 0.5Ω
(3) 1.5Ω (4) 0Ω

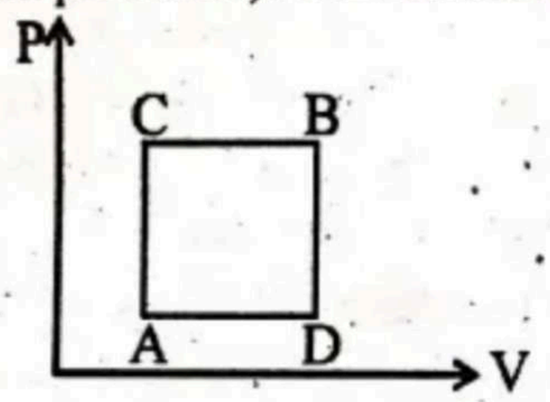
147. A cuboid of dimension 2Lx2LxL, a charge q is placed at centre of surface with area $4L^2$. The flux through the opposite surface to 'S' is given by.
(1) $\frac{q}{12\varepsilon_0}$ (2) $\frac{q}{2\varepsilon_0}$
(3) $\frac{q}{3\varepsilon_0}$ (4) $\frac{q}{6\varepsilon_0}$

148. A solid glass rod of radius 'r' is placed inside a glass cylinder of internal radius 'R'. This glass rod is placed coaxially with the cylinder and the system is held vertically with a part inside the water. Assuming angle of contact 0° and surface tension 'T' then the height to which water will rise in the region between the rod and cylinder is

(1) $\frac{2T}{(R-r)\rho g}$ (2) $\frac{T}{(R+r)\rho g}$

(3) $\frac{2T}{(R^2-r^2)\rho g}$ (4) $\frac{2T}{(R^2+r^2)\rho g}$

149. A gas can be taken from A to B via two different processes, ACB and ADB.



When path ACB is used 60J of heat flows into the system and 30J of work is done by the system. If path ADB is used work done by the system is 10J. The heat flow into the system in path ADB is:

(1) 80J (2) 20J (3) 100J (4) 40J

150. Two liquids A and B are at temperatures of 75^0C and 15^0C. Their masses are in the ratio of 2:3 and their specific heats in the ratio 3:4. The resulting temperature if they are mixed is

(1) 90^0C (2) 70^0C

(3) 35^0C (4) 60^0C

CHEMISTRY

SECTION – A

151. Most electronegative element among the following on Pauling scale is

(1) S (2) P

(3) Br (4) H

152. SF_4, CH_4, ClF_3, SO_3. How many are non-polar molecules.

(1) 0 (2) 2

(3) 1 (4) 3

153. Statement-I : Simple distillation is used to separate volatile liquids from non-volatile impurities having sufficient difference in their boiling points

Statement-II : In thin layer choromatogrphy, ninhydrin is used to detect amino acids

(1) I & II are correct

(2) I is correct, II is incorrect

(3) I is incorrect, II is correct

(4) I and II are incorrect

154. $E^0_{Zn^{2+}/Zn} = -0.76V$; $E^0_{Cu^{2+}/Cu} = +0.34V$

Emf of Daniel cell is

(1) 1.1V (2) -1.1 V

(3) 0.42 V (4) -0.42 V

155. Number of stereo isomers for $\left[Pt(Br)(Cl)(NH_3)(Py)\right]$ and $\left[Pt(en)_2Cl_2\right]^{2+}$ are 'X' and 'Y' respectively. $\frac{X}{Y} =$

(1) 0.666 (2) 1

(3) 1.33 (4) 0.5

156.

Organic compound	Positive test
I) C_6H_5CHO	A) 2,4 – DNP, Tollen's, Fehling 's but not Iodoform
II) $CH_3-CO-CH_3$	B) 2,4-DNP,Haloform but not Tollen's
III) CH_3-CHO	C) 2,4-DNP, Tollen's but not Fehling's
IV) $HCHO$	D) 2,4-DNP,Fehling's and Iodoform

Correct match is

	I	II	III	IV
(1)	A	B	C	D
(2)	C	B	D	A
(3)	C	B	A	A
(4)	A	B	D	C

157. Total number of nodes (angular and radial) for 5f orbital is

(1) 2 (2) 0

(3) 4 (4) 1

158. A salt solution gives white ppt with $NH_4Cl + NH_4OH + H_2S$. If drops of $NaOH$ are added to this salt solution a dirty white ppt is formed which dissolves in excess of NaOH. The cation present in the salt is

(1) Ca^{2+} (2) Fe^{+3}

(3) Zn^{2+} (4) Pb^{2+}

159. Most volatile among the following is

(1) H_2S (2) H_2O

(3) H_2Se (4) H_2Te

160. A catalyst lowers _____of a reaction

(1) ΔH (2) E_a

(3) ΔG (4) ΔU

161. Which one of the following acts like a reducing agent in aqueous solution ?

(1) Eu^{2+} (2) Gd^{3+}

(3) Yb^{3+} (4) Ce^{4+}

162. 2.45 grams of $KClO_3$ on thermal decomposition liberates V ml of oxygen at STP. Value of 'V' is

(1) 6720 (2) 672

(3) 67.2 (4) 67200

163. Which one of the following produces a secondary alcohol ?

(1) $Acetone + RMgX / H_2O$

(2) $Acetaldehyde + RMgX / H_2O$

(3) $Formaldehyde + RMgX / H_2O$

(4) $Hydrogen\ cyanide + RMgX / H_2O$

164. Identify incorrect statement from the following

(1) Melting point of $W > Mo > Cr$

(2) Density of $Cu > Mn$

(3) IP_2 of $Cr > Mn$

(4) Stability of "+6" oxidation state in $Cr > Mo > W$

165. Osmotic pressure of 500 ml solution containing 18 grams of non-volatile solute is 4.926 atm at 300 K. Molar mass of solute is

(1) 180 g/mole (2) 60 g/mole

(3) 200 g/mole (4) 120 g/mole

166. Incorrect statement is

(1) PbF_4 is ionic

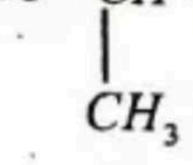
(2) C-14 is a radioactive isotope

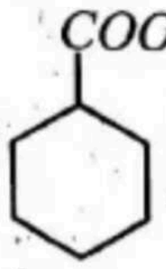
(3) CO is an acidic oxide

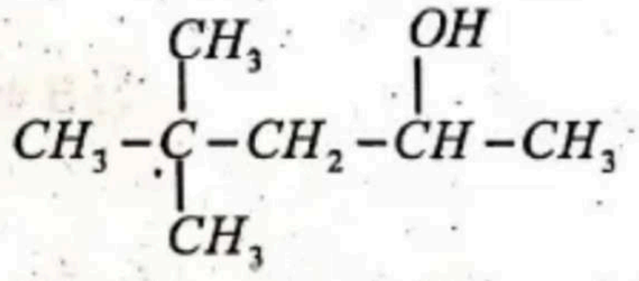
(4) Catenation power of $Ge \approx Sn$

167. Incorrect combination of organic structure and its IUPAC name is

(1) $HC \equiv C - CH_2 - CH = CH_2$ Pent-1-en-4-yne

(2)  ....3-formyl butane nitrile

(3)  ...,Cycloheptanoic acid

(4)  4,4-dimethyl pentan-2-ol

168. According Bohr's theory, correct statements are

A) Angular momentum of electron in 2nd orbit of H-atom is $\frac{h}{\pi}$

B) Velocity of electron is proportional to $\frac{Z}{n}$

C) Radius of first orbit of H-atom is 52.9 A^0

D) Energy difference between 2nd orbit and 1st orbit of H-atom is greater than the energy difference between 3rd orbit and 5th orbit

(1) A, B and D only (2) A,B,C,D

(3) B and D only (4) B, C and D only

169. Which one of the following does not react with Hinsberg reagent.

(1) R_3N (2) R_2NH

(3) RNH_2 (4) $Ph - NH_2$

170. Number of water molecules in 18 ml of water is(density = 1 g /mL)

(1) $\frac{18N_A}{22400}$ (2) N_A

(3) $\frac{N_A}{22400}$ (4) $\frac{18N_A}{22.4}$

171. Ra-226 on radio active disintegration produces α – particles along with
(1) Fr – 225 (2) Fr – 227
(3) Rn – 224 (4) Rn – 222

172. $X \xrightarrow[\Delta]{KMnO_4/H^+}$ Carboxylic acid + Ketone.
'X' is
(1) Propene
(2) 2-methyl-2-butene
(3) 2,3-dimethyl -2-butene
(4) 4-methyl-2-pentene

173. Regarding electron gain enthalpy (magnitude) the correct relation is
A) $S > Se > Te > O$
B) $Cl > Br > F > I$
C) $Ne > Ar > Xe > He$
(1) A and C only (2) B and C only
(3) A, B and C (4) A and B only

174. Mole fraction of solvent in $10\%\left(\frac{w}{w}\right)$ aqueous NaOH is
(1) 0.996 (2) 0.952
(3) 0.927 (4) 0.973

175. Molar conductance of 0.01M weak monoprotic acid (HA) is 40 mho cm^2 mol^{-1}. p^H of the solution is (Limiting molar conductance of HA is 400 mho cm^2 mol^{-1}).
(1) 2 (2) 4
(3) 2.3 (4) 3

176. A salt solution gives a white ppt with $BaCl_2$ solution. The ppt is insoluble in Conc. HCl and also insoluble in Conc. HNO_3. The salt is
(1) NH_4Br (2) FeS
(3) $(NH_4)_2SO_4$ (4) $MgCl_2$

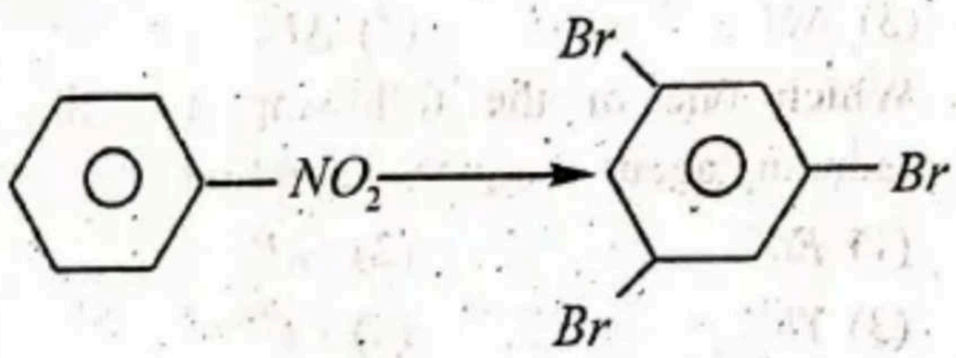
177. Total isomers with the molecular formula, $C_4H_{11}N$ is
(1) 4 (2) 5
(3) 9 (4) 3

178. All the following paris have the same bond order except
(1) N_2^+, N_2^- (2) O_2^+, O_2^-
(3) B_2, O_2^{2-} (4) H_2^+, He_2^-

179. Half life of a first order reaction is $\frac{"x"}{2}$. Time taken for the completion of 87.5% of the same reaction is
(1) 1.5x (2) x
(3) 1.66 x (4) 3x

180. p^{k_a} of CH_3COOH is 4.82. p^{k_b} of NH_4OH is 4.78. p^H of ammonium acetate solution at 298 K is
(1) 6.92 (2) 6.96
(3) 7.02 (4) 7

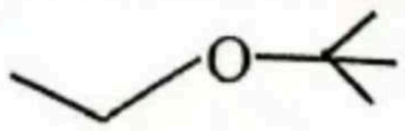
181.



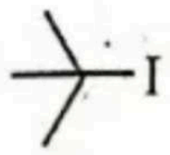
The correct sequence of reagents used for the above conversion are
(1)
$LiAlH_4; Br_2 / H_2O; NaNO_2 + HCl / 278K; H_2O / 283K$
(2)
$LiAlH_4; Br_2 / CS_2; NaNO_2 + HCl / 283K; C_6H_6$
(3)
$Sn + HCl; Br_2 / H_2O; NaNO_2 + HCl / 278K; H_3PO_2 + H_2O$
(4)
$H_2 / Pt; (CH_3CO)_2O; Br_2 / CH_3COOH; NaNO_2 + HCl / 278K; H_2O$

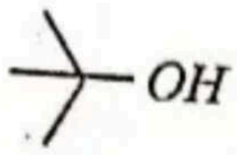
182. $A_3(g) \rightleftharpoons 3A(g)$ is carried out in one litre vessel at 630 K. At equilibrium, number moles of products are twice the number of moles of reactants. Degree of decomposition of $A_3(g)$ is
(1) 0.4 (2) 0.33
(3) 0.25 (4) 0.45

183. Identify incorrect statement.
(1) NF_3 is stable
(2) BiF_3 is ionic
(3) NCl_5 is stable
(4) PF_5 has trigonal bipyramidal geometry

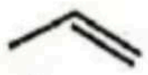
184.  $\xrightarrow[\Delta]{Conc.HI}$ X.

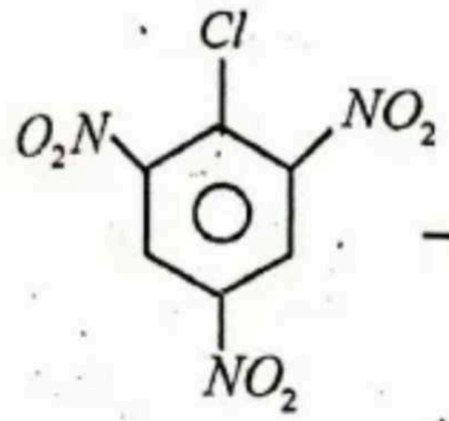
'X' is

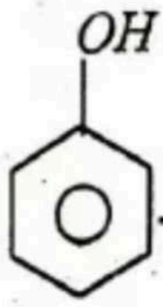
(1) 

(2) 

(3) 

(4) 

185.  $\xrightarrow{Warm\ water}$ A;

 $\xrightarrow[b)Nitration]{a)sulphonation}$ B;

Correct statement is

(1) 'B' is nitro derivative of 'A'

(2) A & B are same compounds

(3) 'A' is less acidic than 'B'

(4) 'A' and 'B' are positional isomers

SECTION - B

186. Negative diviations from Raoult's law are observed in

(1) Chloroform + acetone

(2) Benzene + toluene

(3) n-hexane+n-heptane

(4) Acetone+carbondisulphide

187.

Periodic trend	Property
I) $B > Al > Ga > In > Tl$	A) Ionization potential
II) $B > Al > Tl > In > Ga$	B) Density
III) $B < Al < Ga < In < Tl$	C) Electronegativity
IV) $B > Tl > Ga > Al > In$	D) Boiling point
	E) Melting point

Correct match regarding 13th group elements is

	I	II	III	IV
(1)	E	A	B	C
(2)	A	E	B	C
(3)	A	A	C	B
(4)	D	E	B	A

188. Same element shows more than one type of oxidation state in all the following molecuces except in

(1) $H_2S_2O_7$ (2) Br_3O_8

(3) C_3O_2 (4) $CaOCl_2$

189. p^{k_a} is least for

(1) $BrCH_2CH_2COOH$ (2) C_6H_5COOH

(3) $HCOOH$ (4) $ClCH_2-COOH$

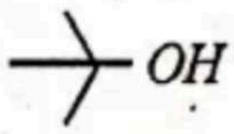
190. Solubility product of AgCl is $1.96\times10^{-10}M^2$. Weight of AgCl dissolved in 696.8 ml of saturated solution is $x\times10^{-4}g$. Value of 'x' is (nearest integer) (GFW of AgCl = 143.5g)

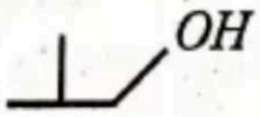
(1) 14 (2) 18

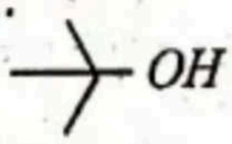
(3) 21 (4) 24

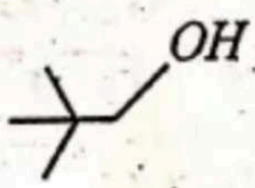
191. $C_2H_5Br \xrightarrow{Na/dryether} A \xrightarrow[anhydrous\ AlCl_3]{HCl} B \xrightarrow{KMnO_4} C.$

Regarding 'C' correct statement is

(1) 'C' is  and gives a ketone with Cu/573K.

(2) 'C' is  and gives an aldehyde with PCC

(3) 'C' is  and gives immediate turbidity with HCl + anhydrous $ZnCl_2$

(4) 'C' is  and gives a carboxylic acid containing four carbons with $KMnO_4 / H^+$

192. Assertion : Melting of ice is spontaneous above 273K.

Reason : Above 273K, for the process of melting of ice, magnitude of $\Delta H > T\Delta S$

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

193. Assertion : Acetanilide on bromination in presence of ethanoic acid gives p-bromo acetanilide as major product

Reason : $-CONHCH_3$ group on benzene ring is an o, p-directing towards electrophilic substitution reactions.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

194. Assertion : Alkaline hydrolysis of methyl bromide gives a product with inversion in configuration

Reason : Alkaline hydrolysis of methyl bromide follows S_N2 mechanism.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

195.

Reducing agent	Reaction
I) $LiAlH_4$	A) $-CO-CH_2-CN \rightarrow -CO-CH_2-CHO$
II) $NaBH_4$	B) $-NO_2 \rightarrow NH_2$
III) $DiBAL-H$	C) $-COOH \rightarrow -CH_2OH$
IV) $Sn+HCl$	D) $-CO-CH_2-COOR \rightarrow -CHOH-CH_2-COOR$

Correct match is

	I	II	III	IV
(1)	D	C	A	B
(2)	C	A	D	B
(3)	C	D	A	B
(4)	D	A	B	C

196. A) $\left[Co(CN)_6\right]^{-3}$ B) $\left[Co(H_2O)_6\right]^{+3}$

Correct combination about the hybridization of central metal in these complexes is

(1) sp^3d^2, sp^3d^2 (2) d^2sp^3, d^2sp^3

(3) sp^3d^2, d^2sp^3 (4) d^2sp^3, sp^3d^2

197.

$$CH_2Br-CH_2Br \xrightarrow{NaNH_2/\Delta} A \xrightarrow[60^oC]{dil.\ H_2SO_4, HgSO_4} B \xrightarrow{Zn/Hg/Conc.HCl} C \xrightarrow[(CH_3COO)_2Mn]{O_3} D.$$

$$A \xrightarrow{X} C \ ; \ B \xrightarrow{Y} D$$

X and Y are respectively

(1) $H_2 / Pd-BaSO_4; PCC$

(2) $H_2 / Ni, \Delta; \left[Ag(NH_3)_2\right]^+ / H^+$

(3) $H_2 / Pt, 298K; NaOI / H^+$

(4) $LiAlH_4; CrO_2Cl_2 + CS_2 / H_3O^+$

198. Greatest tendency to convert from $+2 \rightarrow +3$ in aqueous solution is for

(1) Cr (2) Co

(3) Mn (4) Fe

199. In Nickel-cadmium battery, the change in oxidatioin state at anode and cathode is

(1) $0 \rightarrow +2 ; +3 \rightarrow +2$

(2) $+1 \rightarrow +3 ; +6 \rightarrow +2$

(3) $+2 \rightarrow +3 ; +3 \rightarrow +2$

(4) $+2 \rightarrow +3 ; +4 \rightarrow +2$

200. In 17th group oxides, the correct order of stability is represented as

(1) $Cl > I > Br$ (2) $Cl > Br > I$

(3) $I > Br > Cl$ (4) $I > Cl > Br$